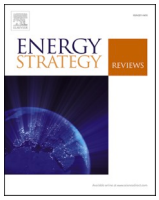
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Barriers to the uptake of solar-powered irrigation by smallholder farmers in sub-saharan Africa: A review

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ABSTRACT

Irrigation expansion is critical for agricultural and rural development, food and nutrition security, and climate change adaptation in Sub-Saharan Africa (SSA). Lack of accessible energy for irrigation development due to limited off- and on-grid infrastructure and the resulting dependence on costly fuel-based irrigation have been key inhibiting factors for irrigation expansion in the region. Off-grid solar-powered irrigation pumps (SIPs) can overcome many of the energy access and other challenges in the region, but their uptake has been slow. Given the nascent development of the solar irrigation sector in SSA, this paper combines a review of the peer-reviewed and grey literature with key informant interviews to identify systemic barriers to the adoption and growth of solar-powered irrigation. We identify uncovered risks, lack of incentives, and lack of capacity as the key factors limiting the adoption of solar-powered irrigation. Moreover, despite significant global cost reductions, solar-powered irrigation systems remain costly in SSA due to limited market development and geographical constraints. Lack of regulation and low investment in building local institutions and value chains further affect uptake and inhibit leveraging the energy transition for ensuring food security and agriculture-led poverty alleviation in SSA. We propose a move away from thinking of SIPs as "silver bullets" and towards a systems approach and the design of context-specific solutions to address risks, incentives and capacity challenges.

1. Introduction

Irrigation development is acknowledged as one of the key strategies for building climate resilience in rainfed agriculture, given increasing climate variability and change in Sub-Saharan Africa (SSA) [1]. Despite the well-documented benefits of irrigated agriculture, including poverty reduction, enhanced food and nutrition security, and improved agricultural productivity [2], its development in SSA has remained low, at around 6% of total cultivated area [3]. Recent estimates show that SSA uses just 1% of its irrigation potential despite substantial scope for expansion [4]. Irrigation development continues to focus on small, medium, and large irrigation schemes that require substantial infrastructural investments. Documented reasons for the slow expansion of irrigation include the high cost of irrigation development [2], the absence of institutional and policy support [5], unclear land tenure, which reduces incentives for structural investments [6,7], poor market linkages [8,9], and lack of financial capacity for more intensive, irrigation-supported agriculture [10,11]. Farmer-led investment in self-owned irrigation technologies provides a viable alternative to the more traditional government- or nongovernmental organization-supported irrigation development, as it overcomes some of the above-identified barriers to slow expansion. However, depending on the water resource, energy access is crucial to enabling farmer-led investments in irrigation.

The role of accessible and affordable energy for agricultural transformation, including irrigation development, is now increasingly

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acknowledged [12]. Unlike energy use for domestic or even industrial purposes, energy needs in irrigation are limited to specific times in a year and depend on the crop, climate, and season. Extending electric grids to rural SSA has been a considerable challenge given the large expenditures associated with laying grid lines in sparsely populated regions [13]. These challenges are compounded for electricity use in irrigation, as areas suitable for irrigation development do not necessarily overlap with rural towns. It is therefore not surprising that electrification can reduce social welfare because the cost of providing electricity exceeds its value given its low usage, as found in Kenya [14]. Despite this, most studies find significant, multiple development impacts from electrification, sometimes not immediate but in the medium to long term, including for health, education, livelihood opportunities, and quality of living [15,16].

Given the large cost of grid infrastructure, the rapid reduction in the cost of solar panels, and the recognition of the climate mitigation benefits of solar systems compared to other energy sources, SIPs are emerging as a promising solution, particularly for small-scale irrigation development [17–19]. Nevertheless, adoption of solar-powered irrigation pumps (SIPs) has remained slow despite their attractiveness over fuel pumps [11] and in the absence of grid electricity [7]. In this paper we investigate the structural reasons that have limited the growth of SIPs in SSA through a multilevel methodological approach explained in the next section. Apart from identification of systemic barriers to the adoption and expansion of solar-powered irrigation, we identify the interlinkages between the barriers to locate potential entry points for addressing these constraints. Some barriers may not be unique to SIPs but are instead linked to the structural ecosystems that embed rural areas and the overall economies of SSA. Hence, it is important to note that the majority of these barriers are not necessarily intrinsic to solar irrigation systems but would be applicable to any new technology or technology-driven solution in the region. This paper contributes to the previous literature on solar irrigation by moving beyond the small case-study approach to applying a more formalized way to assess constraints and by considering the structural dimensions of barriers across the entire SSA region.

2. Methodology and analytical framework

Given the varied hydrological, economic, and political contexts across the entirety of SSA, we adopted a multilevel methodology to collect primary and secondary data in three SSA regions. We started with a literature review of solar-powered irrigation in SSA, which provided us with a bird's eye view of the status of and challenges associated with this technology. Based on the review we selected six countries for a deeper review and analysis in the eastern, western, and southern regions of SSA, namely: Ghana and Mali in West Africa, Ethiopia and Kenya in East Africa, and South Africa and Zimbabwe in Southern Africa. The central region was not covered in this analysis because not much literature was found on the role and uptake of solar irrigation there.

The regional review was followed by regional stakeholder interviews to corroborate our results and gather additional insights. This paper presents the synthesis of the findings and insights from the multilevel literature reviews and key informant (also referred to as stakeholder) interviews.

2.1. Study methodology

Our narrative analysis is based on a review of the peer-reviewed literature using a systematic search. However, given that deployment of SIPs remains very limited and that many development agencies are engaged in case study and implementation work, we also explored a second, substantial body of grey literature to develop insights on solar-powered irrigation. Hence, we complemented the academic literature review with a review of the grey literature. The review methodology follows the guidelines for systematic searches [20] and for conducting an integrative review of the literature [20]. Fig. 1 presents the results of the filtering process we adopted.

2.2. Search strategy

As solar-powered irrigation is a relatively recent development, we did not anticipate much published literature from before 2000. Therefore, we included peer reviewed and grey literature published between 2000 and 2021 in English language. Additional papers were added via "snowballing" sources using the reference lists of selected articles. The following search terms were used to identify articles that contained the specified words in their title, abstract, or keywords, with minor modifications to fit the search algorithm of each database. For example:

Publication date >2000 ("solar irrigation") AND ("Sub-Saharan Africa" OR "Eastern Africa" OR "<COUNTRY>")

A broad range of search keywords was applied to title-abstract-key word searches to identify relevant articles. Many variants of words were also used to improve the filtering process. All keywords used are listed in Annex 1. Peer reviewed, scholarly databases used were JSTOR and Scopus. Grey literature was retrieved through searches conducted in the databases of the United Nations Food and Agriculture Organization (FAO), the International Renewable Energy Agency (IRENA), the World Bank Group, the African Development Bank, Energy for Access, and 60 Decibels, among others.

2.3. Study inclusion and exclusion criteria

a) Relevance of subject

As solar-powered irrigation is a relatively new theme in irrigation development in SSA, not many studies and evaluations have been conducted yet. Instead, the literature mostly consists of reports documenting small-scale experiments, pilots, and demonstrations, largely published over the last decade [7]. However small, these studies can provide useful insights and complement articles that directly contribute to understanding barriers to solar irrigation expansion in Africa. Studies related to solar home systems and other solar domestic and agriculture equipment were excluded.

b) Geography

Only studies covering a country or region in SSA were included. Table 1 details the inclusion and exclusion criteria.

2.4. Review of country-level literature

Following the systematic literature, policy document, and case study review, deep dives were implemented at country level that also considered documents available in local languages, but were not necessarily available online. The detailed country studies also considered administrative and government documents.

2.5. Interviews with stakeholders/key informants

To complement the limited set of publications, 55 key informants were interviewed as part of the country deep dives. They included policymakers, solar companies, development agencies, and farmers and were identified from relevant organizations, case study sites, and the literature (Table 2). Following identification of an initial set of key informants, a snowballing technique was applied, whereby additional key informants were proposed by informants who participated in the study.

Informal and semi-structured interviews were conducted to understand the emerging landscape and barriers to the growth of solar irrigation experienced by different stakeholders in five of the six deep dive countries. No interviews were carried out in Mali due to the challenging geopolitical situation at the time of paper writing.

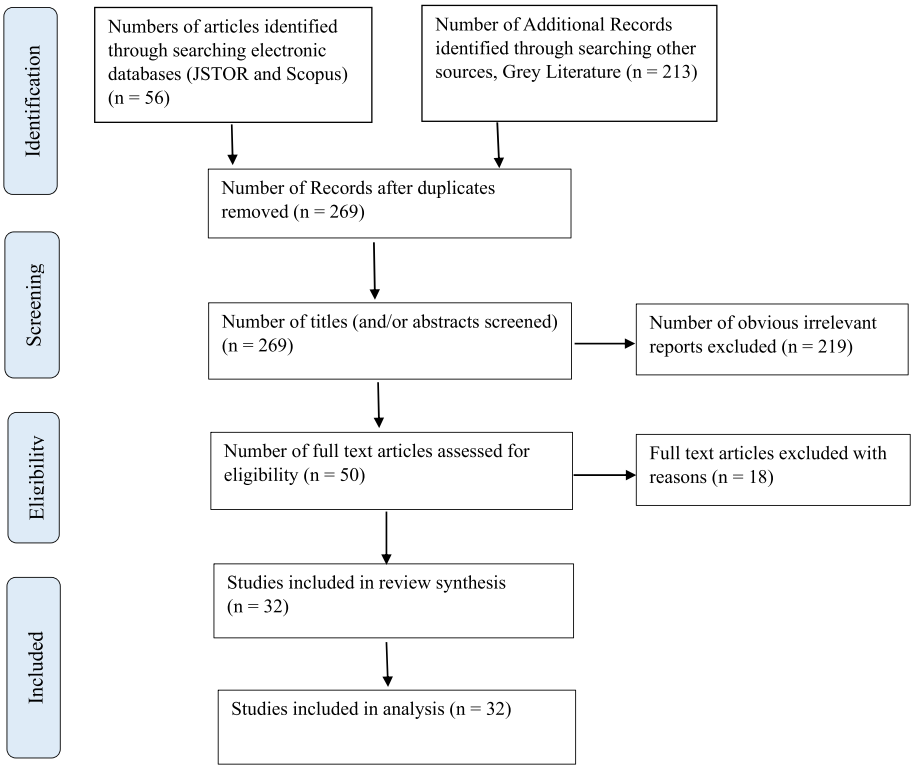


Fig. 1. Selection and filtration of literature for review.

Table 1
Inclusion and exclusion search criteria.

	Include	Exclude
Publication type	Journal articles, working papers, and donor, private sector, and implementer reports assessed by authors for their quality	Abstracts, promotional materials, advertisements, blogs, newspapers
Date of publication	January 2000 to 08/2021[a]	Publications prior to and after the inclusion dates
Geography	Sub-Saharan Africa	All other regions
Language	English	All other languages
Technology	Solar-powered systems for delivering irrigation to smallholder and marginal farmers	Solar home systems, solar panels primarily for domestic electricity generation; solar-powered systems used in large-scale commercial farms

[a] The market price information related to solar pumps was captured from different technical catalogues published in 2022.

2.6. Analytical framework

We adapted the framework laid out in the farmer-led irrigation development guide [21] by customizing the different categories of constraints affecting the uptake of SIPs (themes) and identifying associated subthemes from the literature (Table 3).

3. Identified barriers preventing SIPs from scaling

Table 4 lists the barriers to scaling SIPs identified in the literature, complemented with primary and secondary information collected at country level. While the themes draw on the framework of the farmer-led irrigation development guide [21], the subthemes emerged from the narrative analysis of the literature. These were further elaborated

Table 3
Themes and subthemes of barriers.

Themes	Subthemes
Natural Resources	Aquifer characteristics, water scarcity, irradiance, clouding
Technical	Poor quality standards, obsolete technology
Financial	High capital cost, lack of credit access, financing ecosystem
Market	Lack of relevant products, lack of after sale support services, market infrastructure, and functioning
Policy & Institutional	Lack of long-term structures and institutions for systematic promotion of solar-powered irrigation
	Lack (or ineffectiveness) of policies – trade, import, energy, water, agriculture
	Lack of long-term infrastructure development, poor rural market access
Cultural	Constraints specific to marginal farmers, women, and other socially vulnerable groups that lead to inequitable access and exclusion

Source: Drawing on [21] for themes.

Table 2
Stakeholder interviews.

	Country	Key Informant Interviews With:				Total
		Policymakers	Solar companies	Development agencies	Farmers	
East Africa	Ethiopia	2	4	7		13
	Kenya		3	4		7
West Africa	Ghana	3		6		9
Southern Africa	South Africa	1	11		2	14
	Zimbabwe		6	2	3	11

Table 4
Identification of barriers.

Theme and subtheme	Reference	Southern Africa	East Africa	West Africa
		Zimbabwe, South Africa	Ethiopia, Kenya	Ghana, Mali
1. Natural Resources				
Hydrogeological characteristics of aquifers are not suitable	[10,22–24]	Groundwater availability is limited in the hardrock region of South Africa.	Unavailability of groundwater and expensive drilling of wells.	Aquifer characteristics and groundwater quantities are considered insufficient in many parts of the two countries/region.
Water quality issues (higher iron content in water)	[22,25]	Aridity and dissolution from geological formations hosting the aquifer lead to water quality issues.		Poor quality of groundwater due to salinity and illegal mining.
Energy output from solar panels is low in areas with cloud coverage	[10,11,22, 24,26]	Issue is more pronounced in smaller size pumps because they have very low-capacity panels, unlike bigger pumps where pump HP to kWp of panel ratio is maintained at 1.		
Water scarcity limits the use of SIPs	[6,9,27–30]			
Excessive dust accumulation due to dusty environment	[10,24]			
Risk of floods and climate-related droughts	[31,32]	The effects of climate change may result in reduced rainfall and increased temperatures in some areas, aggravating groundwater depletion and scarcity.		In northern Ghana, destruction of irrigated crops due to floods leads to low income for farmers, which inhibits their financial capability to invest in irrigation infrastructure.
2. Technical				
Repair of solar panels and solar systems is difficult	[6,32–34]			
Susceptibility to lightning strike	[6,29,35]			
Poor product quality	[3,10,25,32, 34,36–38]	Quality of products is good in South Africa, but fake and poor-quality products are a major issue in Zimbabwe. Breakdowns are very frequent in Zimbabwe.		Lack of affordable and high-quality products results in sale of cheaper, low-quality products. In Mali, low-quality SIPs, mainly from China, have a very short lifespan of 2–3 years.
3. Financial				
Higher upfront cost for farmers compared to other pumps	[3,6,7,9–11, 17,23,24,29, 34,36–42]		SIP prices increased in Ethiopia while decreasing globally. Lack of adequate credit deepens the barrier.	Except for low-capacity portable SIPs, most regular-size SIPs has high upfront costs, especially to access groundwater, in comparison to surface water pumps.
Lack of credit facilities and environment for farmers	[7,9,23,25, 28,32–34,38, 43,44]	Zimbabwe lacks credit facilities and products; in South Africa, many options are available for availing credit for larger/commercial farmers, while credit options suitable for small farmers are lacking.	Agriculture is not a viable sector for extending credit; small loans have high interest rates; collateral is needed to access credit from private banks.	Poor access to and very high interest rates in the formal credit system and low capacity of the informal credit system limit farmers' financial capability.
Lack of supportive credit environment for developers and suppliers	[7,33,44]	A few small supply companies in Zimbabwe have started rolling out pay-*as*-you-go systems for small pumps (<1 HP).	Lack of suitable payment mechanisms limits the availability of credit; lack of foreign exchange makes it difficult to import (particularly in Ethiopia).	Both formal banking and microcredit systems are lacking in the region.
4. Market				
Limited availability of suitable or customized products (sizes, types of pumps)	[25,32,34, 38,39]	Lack of appropriate design and capacity of SIPs.	Poor-quality generic products and lack of relevant products inhibit adoption.	Limited understanding and tailoring of businesses to different market segments.
Lack of maintenance and repair services ecosystem	[25,32,34, 36,41]		Lack of support to make the right decisions for sizing and model selection.	Companies prioritize the pre-installation and installation phases over the after-sale phase.
Lack of spare parts	[3,7,25,33]		In Kenya, shortage of spare parts, limited qualified technicians, and limited outlets for accessing pumps inhibit adoption. In Ethiopia, access to spare parts is particularly challenging.	
Low farm density	[2,9]		High cost of logistics.	Cost of creating a distribution network is very high.
5. Policy and Institutional				
Lack of appropriate trade, energy, and water policies and lack of promotion of agriculture and solar-powered irrigation	[7,9,25,34, 41]		The available tax exemption policy in Ethiopia could be viewed as a positive step, but its inconsistency and slow implementation is hindering scaling of SIPs. Restrictive monetary policies limit SIP imports in Ethiopia and increase end-user prices.	Lack of policy and strategies to support irrigated agriculture; lack of policy framework for youth and gender inclusion in solar-based irrigation value chains.
Lack of policy for institutional support for extending finance to farmers and developers	[7,9]	Lack of financial institutions in Zimbabwe.	Microcredit institutions provide loans for agricultural inputs such as agrochemicals, which are much lower than the cost of SIPs in Ethiopia. In	

(*continued on next page*)

Table 4 (*continued*)

Theme and subtheme	Reference	Southern Africa	East Africa	West Africa
		Zimbabwe, South Africa	Ethiopia, Kenya	Ghana, Mali
			Kenya, savings and credit cooperative societies face a similar problem.	
Lack of institutional support to tame theft and vandalism	[6,7,23,25,32]	Key informants cited theft of panels as a major obstacle to adoption of SIPs.		Farmers who invest in solar technology sometimes need to replace their solar panels or abandon their solar pumps due to solar panel theft.
Lack of institutions to train and build capacity of local entrepreneurs, service providers, and technicians	[7,25,32,33]	No mechanisms for training and building capacity for using SIP technology.		Lack of capacity to design cost-effective customized systems. No demonstrations by either state or market.
Lack of coordination among policies, institutions, and agencies	[6,25,37,42]			Institutional pluralism in land policy leads to land-related disputes, making groundwater development financially riskier.
6. Cultural				
Women are significantly more constrained to access information, credit, and services	[23,25,31,32,38]		The constraints faced by women in availing information and techno-financial services are much higher than those faced by men. Even suppliers do not target women.	Most extension workers are men, limiting the information flow to women; government support to women is skewed toward postharvest management aspects.

through key informant interviews and country-specific literature from the deep dives. Some topics—such as dust on solar panels affecting their performance, or general water scarcity affecting uptake of solar-powered irrigation systems—were present in SSA-wide literature but were not emphasized in the literature pertaining to the specific countries/regions.

3.1. Natural resources challenges

The hydrogeological characteristics of aquifers determine the potential of groundwater development both in quantity and quality for irrigation [45], and thus the potential scope of SIPs. SSA is characterized by four major aquifer system types, ranging from hard-rock to alluvial: i) the crystalline basement complex, underlying approximately one-half of the SSA population; ii) consolidated sedimentary rock, which supports close to one-quarter of the SSA population; iii) unconsolidated alluvial sediments supporting around 15% of the population; and iv) volcanic rock supporting around 10% of the population. The crystalline basement complex offers variable yields, the consolidated sedimentary rock provides poor yields, and the unconsolidated alluvial and volcanic rock offer good to very good groundwater yields [45]. Scaling of SIPs is directly linked to aquifer properties and insufficient knowledge of the prevailing hydrogeology can result in inadequate well yields [5,6], saline water intrusion [22] inappropriate SIP selection, and SIP failure. Salt, iron, and other water quality issues also constrain [25] the adoption and expansion of SIPs.

Country-level studies and key informant interviews reflected perceived and real water scarcity, groundwater depletion, and difficulty in extracting adequate groundwater quantities linked to prevailing aquifer characteristics. In parts of Ethiopia and Kenya, the high cost of drilling linked to aquifer characteristics constrains groundwater irrigation and SIP rollout [45], but SIPs are used for accessing surface water. Similarly, hardrock aquifers in South Africa and Zimbabwe are characterized by low storage and low discharge [46]. In some parts of Ghana, groundwater is insufficient for sustainable expansion of SIPs. Salinity is limiting the adoption and expansion of solar irrigation in the Keta Strip and Ga East regions of the country. Water quality challenges linked to mining activities also undermine the beneficial uses of SIPs in part of Ghana [47].

Other challenges affecting SIP scaling in SSA include excessive dust accumulation on panels, cloud coverage, floods, and high temperatures that adversely impact the viability and output from photovoltaic (PV) systems (Table 4).

3.2. Technical constraints

As SIP technologies are relatively nascent, the number of suppliers as well as pre- and after-sale services such as repair and maintenance remain limited SIPs [25]. A further technical constraint identified is lightning, as SIPs are placed at an elevated height [6]; protection from lightning increases system cost. The most pronounced technical barrier to expansion of SIPs is the lack of cost-effective ways for quality assurance of panels and pumps, which results in widespread sales of poor-quality products [10,36].

In Ghana, the availability of low-quality, low-cost SIPs affects widespread adoption and scaling of the technology (KI1).[1] This is compounded by poor sizing of pumps; that is, pumps are often sized too small to extract sufficient water for the irrigated area and crop either because of lack of funds for larger pumps or lack of knowledge on appropriate sizing of pumps vis-à-vis the peak crop water requirement. The latter is compounded by a lack of licensing or accreditation of pump installers [48].

In Zimbabwe also, breakdown of SIPs is linked to low-quality and fake panels (KI2). In South Africa, where the solar irrigation economy is more mature, standardization is available, and breakdowns are less likely due to poor or fake products [3].

3.3. Financial challenges

Several financial challenges arise when it comes to scaling SIPs in SSA. High upfront cost [6,7,9,11,17,34,36,37,39,40] and lack of access to affordable credit for farmers [7,23,25,32,33,43] and for suppliers [7,44] are the most pronounced financial barriers cited in the literature. Unlike in the rest of the world where solar power in general, or solar-powered irrigation specifically, is promoted by governments, in SSA most SIP technologies are directly purchased by farmers from private sector vendors. The poor financing ecosystem for the suppliers hampers market development and availability of innovative financing mechanisms for end-users (for example, PAYGO/PAY-OWN) at scale. Market transactions are constrained due to a lack of finance available to agricultural enterprises and costly informal lending systems. Lack of innovative, customized financial products and models [32] suited to smallholder farmers is a key constraint to the expansion of SIPs in the region and skews access to this technology toward larger, more commercial farms. Even PAYGO or PAY-OWN financing arrangements

[1] Key Stakeholder 1: KI1.

bundled with SIPs require a 20–25 % upfront investment cost, which remains high for smallholder farmers.

A recent estimate suggests that to irrigate 55 million hectares of currently rainfed cultivation, which is close to 40% of total cultivated area (considering 19 crops), an investment of US$3 billion will be needed, which could potentially generate US$5 billion annually [18]. This scale of investment is only possible via innovative business models and long-term planning and deployment through public-private partnerships.

Financial access for smallholder farmers is a key constraint across all three SSA regions. In Zimbabwe, poor access to affordable finance is due to scarcity of financial institutions capable of providing loans to farmers (KI3, KI4). Lack of finance for acquiring solar-powered irrigation is also a key bottleneck in Ethiopia and Kenya. Agriculture is viewed as a high-risk sector and agricultural credit often attracts higher interest rates, even though irrigation reduces risk [49,50]. Moreover, credit levels provided may be insufficient for the acquisition of SIPs [50] or lack adequate collateral [50–52]. Lack of collateral is a constraint particularly faced by women farmers [54]. In some countries, private sector companies use an asset-based financing modality whereby solar pumps are paid back through PAYGO or PAY-OWN models. This eliminates the need of farmers to obtain credit for the entire pump up front and puts the onus of arranging the finance on the supplier or project developer. However, the credit scoring assessments used are often gender-blind, not taking into account off-farm income. Furthermore, solar pump providers themselves face multiple barriers in accessing finance for scaling up pay-go models in SSA (KI5, KI6) as the modality is highly dependent on the company's liquidity.

In South Africa, on the other hand, financial products are available but mostly directed at larger, commercial farmers (KI7) and relevant products suitable for small farmers are rare. The existing agriculture financing structure in South Africa is skewed toward big farmers and would require simplification to make it suitable for small farmers (KI8). However, cooperatives or farmer groups might help groups of smaller farmers to access loans or grants for SIPs from government and other agencies in South Africa [46].

In Ghana, a key informant noted that interventions and projects are skewed toward subsidized technology provision by the international development sector, rather than developing innovative financial products that allow farmers to adopt the technology (KI6, KI9), although some business models are being developed [53]. This further hamper market development as subsidies are not conducive for establishing well-functioning supply chains.

3.4. Market constraints

SIP sales in SSA are largely driven by development agencies [23] as smallholder farmers lack the financial means and SIP markets remain weak. Hence, current products are driven by development agencies and neither meet individual farmer needs [7,28,32] nor stimulate innovative financing arrangements. This along with fuel subsidies [32] prevents farmers from adopting SIPs. Underdeveloped financial markets and markets for other services such as bore hole drilling, agro-input, and off-take provide additional barriers to SIP adoption. Also, the lower density of farmers where solar irrigation is found to be suitable [33,54] results in a higher cost of market development. Smallholder farmers are highly price-sensitive and suppliers face a trade-off between supplying a more customized, durable pump at a higher cost or developing a more generic, modest-quality, affordable pump that may require more maintenance and operational expenditures over time [36].

Lack of an attractive and sizable market results in poor after-sale maintenance services [7,26,28,32] and a scarcity of spare parts [25,33], which result in poor repair and maintenance services of installed SIPs, increasing barriers to their adoption by farmers.

In South Africa, according to some solar pump companies interviewed, the lack of access to market information on solar products, their performance, and warranty conditions makes it difficult for companies to develop the market for solar-powered irrigation (KI10, KI11, KI12). In Ghana and Ethiopia, it is observed that lack of domestic production of solar pumps means that not just the solar pump/PV technology are imported but also the associated products, which adds significantly to the cost. For example, SIPs in Ethiopia were found to be 2.4 times the cost in Kenya. Lack of relevant and customized products for different segments of farmers is another identified factor inhibiting market growth in the region [48].

3.5. Policy and institutional barriers

Lack of political stability in some SSA countries results in lack of long-term policy and institutional alignment [31,41]. And the lack of financial capabilities to support agriculture and irrigation development objectives [9,37] results in many of the currently observed barriers in adoption and scaling of SIPs. Many development agencies support multiple projects in the region with similar objectives but actions are uncoordinated. This scattered approach prevents effective use of financial resources [37] in creating a conducive ecosystem for the development of SIPs. Involvement of multiple financing centers and agencies responsible for design and implementation of projects adds to bureaucratic complexity for technology producers, suppliers, and farmers [37,55]. The lack of well-developed markets and lack of quality assurance standards and benchmarks, result in poor protection of suppliers, developers, financers, and farmers against poor-quality products [9]. Furthermore, lack of capacity building and training programs and institutions—resulting in a lack of trained technical and managerial professionals to support energy transition in irrigation [25,37]—hinder SIP adoption. Also, no regulatory system is in place to ensure quality and standardization of solar pumps in South Africa (KI13), Zimbabwe (KI14), Ghana (KI1), Kenya [56] and Ethiopia [52].

In Ethiopia, shortage of foreign currency is touted as a key constraint preventing the private sector from developing well-established irrigation value chains and financial ecosystems, as technologies and products are imported. Moreover, access to solar pumps and farm inputs lack integrated support systems and are hampered by the National Bank of Ethiopia's regulation on foreign exchange [57]. Despite having zero import duty and value added tax on solar pumps in Ethiopia, their import is often inhibited by lengthy and slow bureaucratic processes. In Ghana, tax exemptions are applied for solar panels only, not for pumps. This complicates the process, as pumps can be used for other than agricultural purposes and hence are not subject to tax exemption (KI15). Furthermore, the tax exemption applied by private sector companies, if successful, will not necessarily result in lowering farmers' investment cost.

In Kenya, SIPs are locally assembled though the parts are imported. However, in many other countries SIP systems are always imported. Lack of domestic production, which requires institutional capital and support, results in reliance on imported products, adding significantly to the cost of the product, inhibiting adoption of the technology in all selected countries.

Finally, lack of technical training for farmers and lack of policies to support youth and women to provide ancillary services associated with solar pumps limit the potential of the technology and inhibits it from scaling (KI5, KI6, KI16, KI17).

3.6. Cultural barriers

Very few papers and information are found on how cultural norms influence adoption of SIPs across Africa. Women's roles in farming in SSA are increasingly being acknowledged [25,31] and "feminization" of agriculture is being observed in many regions due to factors like migration [31]. But compared to men, women continue to face more challenges in accessing information, finance, and services related to agriculture, irrigation, and SIPs [31,32]. It may be noted that no direct

barriers related to sociocultural and gender aspects are observed with respect to adoption and scaling of SIPs. However, sociocultural and gender differences reinforce the already existing barriers. The poor understanding of sociocultural and gender differences in addressing these barriers widens the inequality gap when it comes to SIP access and adoption through development programs and private sector initiatives.

4. Inherited systemic barriers and interlinkages

Several barriers to adoption and scaling of SIPs are inherited from other constituting sectors and hence their resolution may also lie outside the purview of SIP development. We identified several key barriers to adoption and scaling of SIPs that are inherited by the solar irrigation sector.

4.1. Agriculture and irrigation sector

It has been widely identified that poor agriculture and irrigation development owing to a lack of backward and forward linkages and institutional vacuums [5] have deepened existing poverty and risk due to climate change. Agricultural production risk and poorly functioning markets affect the interest and incentives of financial institutions in investing in smallholder agriculture [9]. This is compounded by a lack of collateral assets and viable business plans of many smallholder farming operations [33]. While irrigation of high-value crops using SIPs is proven to be financially viable depending on water source and system capacity [58], most irrigation development supported by the public sector continues to promote low-value staple crops. Hence, even when credit is extended, loan repayment is difficult given the low surplus generated by farmers due to low-value crops and low productivity [34]. This is evident from a primary survey of 400 solar pump users, where nearly one-half of farmers indicated significant challenges with repaying their loan [26]. Other traditional challenges facing farmers that carry over into SIPs include a lack of robust agricultural value chains [9,37] and organized markets, competition from big commercial farms, and conversion from rainfed to irrigated agriculture [17,36,40].

Another important aspect that hinders the adoption and growth of SIPs is the cost associated with groundwater irrigation. The high cost of drilling, poor hydro (geo)logical information, and the low capacity of local drilling services [3] make it difficult for smallholder farmers to invest in wells. Streams and surface water are often not accessible or do not offer a reliable source of water [9] limiting the adoption of SIPs in SSA in the absence of affordable groundwater development. Also, the low efficiency of many pumps is a characteristic of the mechanics of the pumping technology and not specific to SIPs.

4.2. Energy sector

Despite declining prices, solar PV remains the most significant cost element in SIPs. Across the globe, large-scale solar PV power generation is increasingly being adopted [35] resulting in development of a global PV market and support ecosystem [35]. In SSA, a series of structural factors render the energy sector less attractive to renewable investments, such as large-scale solar PV deployment. This includes a lack of clear electrification plans (for example, Kenya and Tanzania), and limited on-site planning, institutional, and financial capacity [35]. Moreover, the lack of trained professionals for enabling the energy transition results in poor development of service markets of ancillary products needed for the energy transition and in poor after-sale maintenance support [26].

Enormous costs associated with grid expansion in sparsely populated areas and lack of models ensuring equitable distribution of electricity subsidies are major barriers in planning and executing electrification programs [13,59]. Even when electricity is extended to rural areas, electricity usage of poor households tends to remain low given the high cost and poor reliability of electricity and the lack of appliances and opportunities to generate income from electricity access. This accentuates the cost of grid expansion and operations to rural regions, where average incomes are low to very low [13]. A recent study found that large-scale initiatives to expand domestic energy access may not produce meaningful economic impacts unless combined with complementary programs that enable low-income groups to access electrical appliances or opportunities to use the energy for economic gains [15]. Similarly, if SIPs are promoted in regions and communities where energy use is negligible or farmers have not realized gains from using irrigation, SIPs—on their own—might not produce the expected results unless coupled with other agricultural services and knowledge products. Therefore, even if SIPs hold the potential to transform rainfed agriculture, a steep learning curve remains if there is a local lack of experience on energy use in agriculture and on irrigation. This would delay the expected results and limit the financial attractiveness of SIPs.

4.3. Solar PV sector

Solar PV is the primary input in decentralized solar applications. SIPs are one such application. The techno-economic development and advancement of solar PV is beyond the solar irrigation space but significantly impact the design, cost, and maintenance of SIPs. Over the years, the cost of solar PV has declined and solar cells have become more energy- and space-efficient. This has contributed to the improved economics of solar PV-based applications, including solar home systems and solar mini-grids.

The techno-political barriers that limit expansion of the solar PV sector in a region also significantly impact the industries using solar PV as an input, including the SIP industry. This was seen recently in South Asia and parts of Europe, where a ban on imports of solar PV from China created a ripple effect on the different sectors using solar PV as an input [60–62]. Other challenges of the solar PV sector that can affect SIPs include scarcity of some source materials and the disposal of solar panels following their useful lifespan.

4.4. Financial sector

Financial institutions have complex modalities and systems to assess risk to operationalize credit disbursement [32,37]. Credit rating systems often reject poor farmers, especially women, as they tend to lack experience and technical capacity to understand such modalities and requirements and because they often lack collateral [7,25,31]. Compared to extending credit in urban areas, the cost of extending credit to marginalized and rural communities is often higher, given their lower population density and the riskier, lower- and less frequently profit-generating enterprises most farmers are engaged in. For rural financial systems to flourish, institutional and policy support is needed [7,37] whilst strengthening the SPIS supply chains. These supply- and demand-side barriers in expanding access to affordable credit in rural areas also affect affordable credit products and services for SIPs in SSA. Furthermore, the wider financing ecosystem to stimulate entrepreneurship of farmers and wider market development of agri-businesses remain weak in many countries in SSA.

4.5. Gender

Women in SSA have lower or no ownership of land [31–33] and other assets than men. Moreover, their mobility is generally reduced compared to that of men, and they often benefit from fewer years of education. As a result of these inequities, their bankability is systematically lower than that of men, and their access to finance is more constrained, thus reinforcing the idea that solar technology could worsen “socio-economic asymmetries” [7]. Even innovative financing arrangements such as PAY-GO, stemming from the solar home systems energy sector, use credit-scoring mechanisms that exclude women. However, incipient attempts to strengthening women’s access to finance for SIPs

do exist [54].

5. Problematic SIP scaling solutions proposed in the literature

In this section we discuss some of the suggestions found in the literature for improving the uptake of SIP that are not borne out by our in-depth review of factors of success in SSA.

5.1. Ancillary additions will improve uptake

It has been suggested that the use of remote operation and the "Internet of Things" (IoT) may improve the product design and hence adoption of SIPs [36,44]. But given that high capital cost is one of the most critical barriers hindering adoption of SIPs, such ancillary features would increase the capital and maintenance costs of the systems and would require additional capacity for repairs that is generally not available in rural areas of SSA. This would reduce their affordability further, hindering their adoption and scaling. However, such ancillary features may add value in ensuring that operation and repair services are available and provided, research pilots and experiments to improve product design, customize systems, or understand the behavior of irrigators.

5.2. Small pumps are economical

The other idea widely presented in the literature is that smaller PV systems solar pumps are more affordable. Observing the trajectory of solar-powered irrigation expansion in more mature markets we find that the smallest systems (<2 HP) are more expensive than slightly larger (3–5 HP) systems if we compare cost per kWp (Fig. 2). Table 5 compares the market prices of different pump sizes in the selected countries. We observe the same trend: smaller pumps tend to be more expensive and the marginal cost for acquiring a slightly larger-sized pump declines with increased pump size. We see a significant reduction in the marginal cost of adding a kWp in Mali. But the same is not true for Zimbabwe, where the marginal cost per kWp addition only changes slightly. Stakeholder interviews revealed that in the SSA region, SIP prices are highly dependent on the country's import duty on Chinese products. In countries like Zimbabwe, where importation of Chinese products, including SIPs, is relatively easier, SIP prices tend to be lower and a larger variety of pump sizes is available. The price also varies by brand: brands produced in Europe tend to be more expensive than lesser-known brands such as Terrier (used in Zimbabwe). A more robust comparison of the usability of smaller and larger pumps across different countries should involve the same type or model of pump, as its technical characteristics significantly impact its performance and price and, hence, the economics of SIPs.

Table 5
Market prices of AC submersible pump in Mali, South Africa, Ghana, and Zimbabwe.

Size of pump (HP)	Price of a pump (US$)			
	Mali	South Africa	Ghana	Zimbabwe
1	6772	2575	1327	489
2	7624	2857	2076	671
3	9761	4703	3797	1150
5	11,301	5095	5423	1646
7.5	14,988	5489	7531	1720
10	17,679	7133	9691	2444

(Sources ([66]for Mali [67,68]; for Zimbabwe; Authors' estimates based on online catalogs of Pumptech for Ghana; Catalogue of Lorentz Pumps for South Africa)

Experimenting with business models with slightly bigger size pumps may reveal useful insights for scaling up solar irrigation in the region. An experiment in Rwanda [63] found that slightly larger pumps were more effective in cost and operational capability in drawing water from deeper depths and delivering at longer distances. Some experiments in India also revealed that larger pumps had better utilization and demand than smaller individual pumps because of their improved economics and performance in the summer season [64,65]. However, comparisons with India need to be treated with caution as India provides high level of subsidies for larger pumps, discouraging production and sales of smaller-sized pumps. Moreover, many other factors—such as farm size, types of crops, water source, and density of farmers—determine the viability of larger-sized SIPs and the uptake of solar or other energy systems, in general.

For example, in Kenya, the benefits of electrification have been found to be highly heterogeneous: better-off households valued electricity more than low-income households, essentially because of the difference in their abilities to leverage energy for improving their income and livelihoods [15]. We expect similar heterogeneity in the costs and benefits accruing to different SIP sizes. Careful curation of programs to suit farmers' requirements can enhance the adoption and scalability of SIP-led irrigation development. This can be supported through the establishment of service-based models alongside ownership models that would support pump rental to overcome challenges with purchase

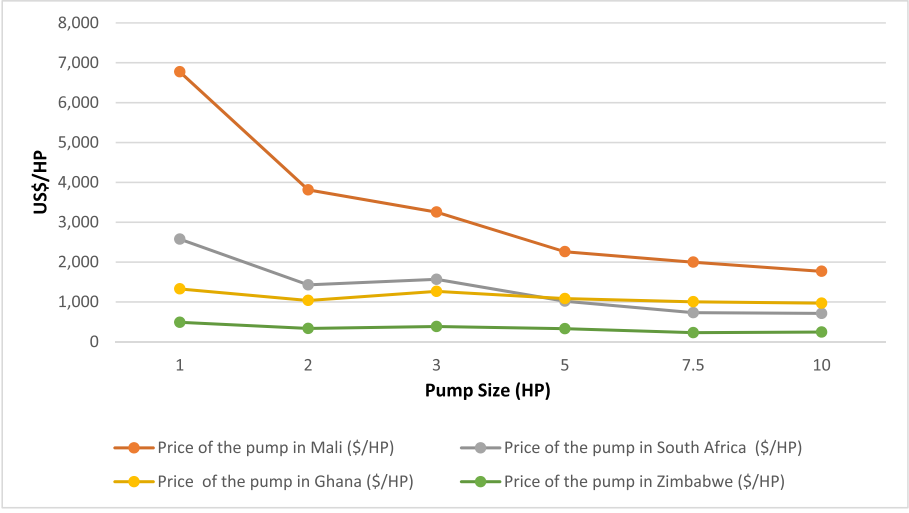


Fig. 2. Size vs. price variation for AC submersible solar pump.
(Source: Authors' analysis)

capacities and could provide other needs-based support to farmers interested in adopting SIPs.

5.3. Groundwater depletion in SSA

A growing debate exists around the perception of water shortages in SSA and how this discourse has led to neglect of much-needed groundwater development, which can greatly contribute to achieving water and food security goals in the region [69,70]. This debate is linked to SIPs with the assumption that solar-powered pumps, which have no variable water extraction costs, will worsen groundwater depletion in SSA [7,32,37]. A survey of 400 farmers conducted in 2019 found that close to one-third of respondents reported water quantity and availability issues, but few assessments indicated that overextraction is not at all a near-term issue [26]. A recent calculation by researchers suggests that SSA has used just 1% of its irrigation potential and significant scope remains for sustainable irrigation development for improving food and nutrition security and livelihoods in the region [4]. Another study estimated a blue water demand of 67 km^3/year over smallholder-farmed rainfed cropland (considering 19 crop types), which translates into the installation of 11 million SIPs that can be safely deployed without any risk to groundwater overexploitation [18]. However, the study remains unclear on how surface water pumping was modelled. However, groundwater depletion has been identified in localized areas of SSA [48], and hence approaches to scaling SPIS in areas with large renewable groundwater reserves will need to be treated differently than when groundwater irrigation is developed in already water-scarce areas.

6. Interlinkages of barriers

The barriers identified throughout this study are strongly interlinked through different mechanisms. The study identified three principles which explain key interlinkages of the identified barriers.

i) Perceived and real risks for undertaking action
ii) Lack of incentives for different stakeholders to change the current course of action
iii) Lack of capacity of different stakeholders to execute effective decision making

We find that policymakers' lack of capacity to design effective policies results in weak institutions and therefore ineffective incentives for the technical and financial industry to innovate and improve the affordability of SIPs. This in turn influences the perceived and real risks for entrepreneurs and businesses to invest in market development and for farmers to invest in new technology. Furthermore, the perceived and real risks associated with natural resources (especially groundwater overexploitation) with SIP expansion disincentivizes multilateral institutions to significantly invest in SIP-led groundwater development. Table 6 presents these three key themes and the associated barriers.

6.1. Real and perceived risks

SSA's diverse natural resource characteristics demand customization of SIPs to suit pumping requirements, but this results in higher costs of designing and difficulty in standardizing products. Natural resource characteristics such as aquifer type and high salt and mineral content in groundwater may require specialized pumps with high maintenance costs, increasing the cost of SIPs. To enable night time irrigation, batteries and other associated systems also increase the cost of SIPs. Even though, some solutions have been proposed to enable night time irrigation without using batteries such as connecting SIPs to grid but any additional feature increases the capital and maintenance cost of the system. Compromised affordability would deepen the financial barrier for accessing SIPs by smallholder farmers. And customizing products require flexible policies associated with their standardization. Flexible policies, in turn, can be ineffective in ensuring product quality. Lack of protection of suppliers and farmers against low-quality products increases the perceived and real risk of adoption by farmers and other stakeholders. Further, low demand does not incentivize suppliers to import a variety of SIP models nor to develop a variety of financial products. This may lead to only partial replacement of the existing fuel pumps in SSA [39] and might limit expansion of new irrigated areas [26,39].

6.2. Lack of incentives

Lack of policy coherence de-incentivizes suppliers to import high quality pumps at affordable prices and financial institutions to extend improved credit terms to suppliers and farmers. This compromises the

Table 6
Principle themes and interlinked barriers.

Theme	Subtheme	Real and perceived risks	Lack of incentives	Lack of capacity
Natural Resources	Hydrogeological characteristics of aquifer are not suitable	●		
	Water quality issues (higher iron content in water)	●		
	Energy output from solar panels is low in cloudy weather			
	Water scarcity limits the use of SIPs	●		
	Excessive dust accumulation due to dusty environment			●
	Risk of floods and climate-related droughts	●		
Technical	Repair of solar panel is difficult			●
	Susceptible to lightning strike	●		
	Poor product quality		●	●
Financial	Higher upfront cost for farmers		●	
	Lack of credit facilities and environment for farmers		●	●
	Lack of supportive credit environment for developers and suppliers		●	●
Market	Limited availability of suitable or customized products (sizes, types of pumps)		●	
	Lack of maintenance and repair services ecosystem		●	●
	Lack of spare parts		●	
	Lower density of farmers			
Policy and Institutional	Lack of effective policies for trade, energy, and water and promotion of agriculture and solar-powered irrigation		●	●
	Lack of policy for institutional support for extending finance to farmers and developers		●	●
	Lack of institutional support to tame theft and vandalism			●
	Lack of institutions to train and build capacity of local entrepreneurs, service providers, and technicians			●
	Lack of coordination among different policies, institutions, and agencies		●	●
Cultural	Women are significantly more constrained to access information, credit, and services		●	●

size of the serviceable market which makes it unattractive for suppliers to strengthen their supply chains and service models in rural areas. Low serviceable markets for irrigation and specifically SIPs [36] was found to be one of the major barriers to market innovation and growth in SSA [36]: out of a total 600 million farmers, only 6% might be able to afford SIPs, considering individual SIP ownership models [36]. However, service-based leasing modalities or sharing arrangements could significantly increase the serviceable market. Lower demand also disincentivizes market innovation, as described earlier. The lack of developed SIP markets and associated financing arrangements makes it even more difficult for marginalized communities to access new technology, deepening their exclusion.

6.3. Lack of capacity

Lack of capacity extends across all themes assessed in this study. Governments lack capacity and political will to develop, implement and enforce groundwater policies, institutions and regulations that would support expensive investments, such as those in SIPs. Without such policies and action, including active groundwater management in areas where water tables are declining, SIP investment is a lose-lose proposition. Similarly, there are capacity gaps in understanding natural resource constraints to the development of SIPs and other irrigation with systems often developed in the most water scarce and least sustainable regions. Perceived risks and lack of incentives interplay with lack of incentives to reduce the size and cost of SIPs. This also impacts the quality of SIP value chains, finances, and technical support. And even if SIP costs would be reduced by tackling the risks and enhancing incentives, smallholder farmers lack the capacity to develop viable business models and to keep financial records. This makes it difficult for financial institutions and suppliers to assess their creditworthiness [71], and for banks to extend credit to them. In countries like South Africa, where banking services are well organized for commercial farmers, smallholder farmers still struggle to access credit from the formal banking system. Furthermore, even PAYGO or PAYOWN financing arrangements require creditworthiness assessments of smallholder farmers. Given the complexity of irrigated agriculture, suppliers and third-party financial organizations offering these innovative financing arrangements often lack the capacity to assess the creditworthiness and estimate default rates of farmers. Therefore, institutions need to strengthen capacity of suppliers and other public and private sector actors to innovate products and procedures to accelerate smallholder farmers' access to affordable financing schemes or service arrangements.

7. Conclusions

This paper focused on understanding the nature of and interlinkages between the different barriers inhibiting adoption of solar-powered irrigation in SSA. We do not attempt to provide any solutions for overcoming these barriers given the diversity and contextuality of problems in the region. However, we do provide a structural analysis of barriers that can be used to identify entry points for potential solutions that addresses the root causes preventing SIP from scaling. This requires systems thinking and transformation of the wider ecosystem by designing context-specific solutions and moving away from thinking of SIPs as "silver bullets" to irrigation development. A thorough understanding is needed of the enabling environment, the demand for and supply of SIPs, and their suitability to the physical nature of the environment. The deep dives focus on barriers experienced in countries with some SIP activity. Barriers are likely larger in other countries that currently have even less SIP development. To address the constraints in both cases, more research is needed on the determinants of uptake, the impact of innovative financing mechanisms on SIP adoption by farmers and ability of suppliers and financing institutions to scale SIP.

While SIPs may be a promising solution for ensuring food security and sustainably alleviating poverty in SSA, a series of interlinked barriers need to be considered in the design of effective scaling strategies. The lack of strong governance and institutions, long-term policy, and budgetary capabilities puts the onus of developing and scaling solutions on market-led mechanisms. Given the heterogeneity of benefits, costs, and the trajectory of adoption associated with SIPs, a curation of context-specific business and inclusive financial models becomes particularly important for developing SIP markets. Given the complex interlinkages across barriers, addressing individual challenges is unlikely to unleash a SIP revolution in SSA. Instead, a package or systems approach is needed that covers several of the main constraints that affect different regions to varying degrees.

CRediT authorship contribution statement

Neha Durga: Methodology, Writing - original draft, Formal analysis, Revision. **Petra Schmitter:** Writing - review & editing, Revision. **Claudia Ringler:** Conceptualization, Writing - review & editing, Revision. **Shrishti Mishra:** Data curation, Formal analysis. **Manuel S. Magombeyi:** Resources, Data curation. **Abena Ofosu:** Resources, Data curation. **Paul Pavelic:** Supervision. **Fitsum Hagos:** Supervision. **Dagmawi Melaku:** Data collection. **Shilp Verma:** Writing - review & editing. **Thai Minh:** Supervision. **Chamunorwa Matambo:** Data collection.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Annex 1.

Search Strings: *("solar irrigation") AND ("Africa" OR "<COUNTRY>")*

Search Terms.

1. Theme
   - "Solar irrigation pump"
   - "Solar powered irrigation pump"
   - "Solar pump"
   - "Solar powered irrigation system"
   - "Solar PV irrigation pump"
   - "Solar PV pump"
   - "Solar PV powered irrigation"
   - "Solar PV irrigation"
   - "Solar PV irrigation system"
2. Region
   - "Africa"
   - "Sub Saharan Africa"

- "Middle East and North Africa"
- "MENA"
- "North Africa"
- "East Africa"
- "Eastern Africa"
- "West Africa"
- "Western Africa"
- "Southern Africa"

3. Countries
   - North Africa: Egypt, Morocco, Tunisia
   - East Africa: Ethiopia, Kenya
   - West Africa: Ghana, Mali
   - Southern Africa: South Africa, Zimbabwe

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